

Import and Process Data

Keman Xiang

Introduction

As a data analyst, it will be common for us to import data from external files into our R console and use it to create a data frame to analyze it. In this notebook we will import data from outside of R using the `read_csv()` function. Then, we will use R functions to manipulate and interact with that data.

The scenario

In this scenario, we are junior data analysts working for a hotel booking company. We have been asked to clean a .csv file that was created after querying a database to combine two different tables from different hotels. In order to learn more about this data, we are going to need to use functions to preview the data's structure, including its columns and rows. We will also need to use basic cleaning functions to prepare this data for analysis.

Step 1: Load packages

Start by installing our required package.

```
#install.packages("tidyverse")
```

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.4
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2    3.4.4      ✓ tibble     3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr      1.3.0
## ✓ purrr      1.0.2
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Step 2: Import data

The data in this example is originally from the article Hotel Booking Demand Datasets

(<https://www.sciencedirect.com/science/article/pii/S2352340918315191>

(<https://www.sciencedirect.com/science/article/pii/S2352340918315191>)), written by Nuno Antonio, Ana Almeida, and Luis Nunes for Data in Brief, Volume 22, February 2019.

The data was downloaded and cleaned by Thomas Mock and Antoine Bichat for #TidyTuesday during the week of February 11th, 2020 (<https://github.com/rfordatascience/tidytuesday/blob/master/data/2020/2020-02-11/readme.md> (<https://github.com/rfordatascience/tidytuesday/blob/master/data/2020/2020-02-11/readme.md>)).

Learn more about the dataset here: <https://www.kaggle.com/jessemostipak/hotel-booking-demand> (<https://www.kaggle.com/jessemostipak/hotel-booking-demand>)

Use the `read_csv()` function to import data from a .csv in the project folder called "hotel_bookings.csv" and save it as a data frame called `bookings_df`:

```
bookings_df <- read_csv("hotel_bookings.csv")
```

```
## Rows: 119390 Columns: 32
## — Column specification —————
## Delimiter: ","
## chr  (13): hotel, arrival_date_month, meal, country, market_segment, distrib...
## dbl  (18): is_canceled, lead_time, arrival_date_year, arrival_date_week_numb...
## date  (1): reservation_status_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Now that we have the `bookings_df`, we can work with it using all of the R functions.

Step 3: Inspect & clean data

Preview the data using `head()` function, which returns the columns and first several rows of data.

```
head(bookings_df)
```

```
## # A tibble: 6 × 32
##   hotel      is_canceled lead_time arrival_date_year arrival_date_month
##   <chr>          <dbl>    <dbl>          <dbl> <chr>
## 1 Resort Hotel          0      342          2015 July
## 2 Resort Hotel          0      737          2015 July
## 3 Resort Hotel          0        7          2015 July
## 4 Resort Hotel          0       13          2015 July
## 5 Resort Hotel          0       14          2015 July
## 6 Resort Hotel          0       14          2015 July
## # i 27 more variables: arrival_date_week_number <dbl>,
## #   arrival_date_day_of_month <dbl>, stays_in_weekend_nights <dbl>,
## #   stays_in_week_nights <dbl>, adults <dbl>, children <dbl>, babies <dbl>,
## #   meal <chr>, country <chr>, market_segment <chr>,
## #   distribution_channel <chr>, is_repeated_guest <dbl>,
## #   previous_cancellations <dbl>, previous_bookings_not_canceled <dbl>,
## #   reserved_room_type <chr>, assigned_room_type <chr>, ...
```

Check out the `str()` function:

```
str(bookings_df)
```

```

## spc_tbl_ [119,390 × 32] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ hotel : chr [1:119390] "Resort Hotel" "Resort Hotel" "Reso
rt Hotel" "Resort Hotel" ...
## $ is_canceled : num [1:119390] 0 0 0 0 0 0 0 0 1 1 ...
## $ lead_time : num [1:119390] 342 737 7 13 14 14 0 9 85 75 ...
## $ arrival_date_year : num [1:119390] 2015 2015 2015 2015 2015 ...
## $ arrival_date_month : chr [1:119390] "July" "July" "July" "July" ...
## $ arrival_date_week_number : num [1:119390] 27 27 27 27 27 27 27 27 27 ...
## $ arrival_date_day_of_month : num [1:119390] 1 1 1 1 1 1 1 1 1 ...
## $ stays_in_weekend_nights : num [1:119390] 0 0 0 0 0 0 0 0 0 ...
## $ stays_in_week_nights : num [1:119390] 0 0 1 1 2 2 2 2 3 3 ...
## $ adults : num [1:119390] 2 2 1 1 2 2 2 2 2 2 ...
## $ children : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
## $ babies : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
## $ meal : chr [1:119390] "BB" "BB" "BB" "BB" ...
## $ country : chr [1:119390] "PRT" "PRT" "GBR" "GBR" ...
## $ market_segment : chr [1:119390] "Direct" "Direct" "Direct" "Corpora
te" ...
## $ distribution_channel : chr [1:119390] "Direct" "Direct" "Direct" "Corpora
te" ...
## $ is_repeated_guest : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
## $ previous_cancellations : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
## $ previous_bookings_not_canceled : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
## $ reserved_room_type : chr [1:119390] "C" "C" "A" "A" ...
## $ assigned_room_type : chr [1:119390] "C" "C" "C" "A" ...
## $ booking_changes : num [1:119390] 3 4 0 0 0 0 0 0 0 0 ...
## $ deposit_type : chr [1:119390] "No Deposit" "No Deposit" "No Depos
it" "No Deposit" ...
## $ agent : chr [1:119390] "NULL" "NULL" "NULL" "304" ...
## $ company : chr [1:119390] "NULL" "NULL" "NULL" "NULL" ...
## $ days_in_waiting_list : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
## $ customer_type : chr [1:119390] "Transient" "Transient" "Transient"
"Transient" ...
## $ adr : num [1:119390] 0 0 75 75 98 ...
## $ required_car_parking_spaces : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
## $ total_of_special_requests : num [1:119390] 0 0 0 0 1 1 0 1 1 0 ...
## $ reservation_status : chr [1:119390] "Check-Out" "Check-Out" "Check-Out"
"Check-Out" ...
## $ reservation_status_date : Date[1:119390], format: "2015-07-01" "2015-07-01"
...
## - attr(*, "spec")=
## .. cols(
## .. hotel = col_character(),
## .. is_canceled = col_double(),
## .. lead_time = col_double(),
## .. arrival_date_year = col_double(),
## .. arrival_date_month = col_character(),
## .. arrival_date_week_number = col_double(),
## .. arrival_date_day_of_month = col_double(),
## .. stays_in_weekend_nights = col_double(),
## .. stays_in_week_nights = col_double(),
## .. adults = col_double(),

```

```
## .. children = col_double(),
## .. babies = col_double(),
## .. meal = col_character(),
## .. country = col_character(),
## .. market_segment = col_character(),
## .. distribution_channel = col_character(),
## .. is_repeated_guest = col_double(),
## .. previous_cancellations = col_double(),
## .. previous_bookings_not_canceled = col_double(),
## .. reserved_room_type = col_character(),
## .. assigned_room_type = col_character(),
## .. booking_changes = col_double(),
## .. deposit_type = col_character(),
## .. agent = col_character(),
## .. company = col_character(),
## .. days_in_waiting_list = col_double(),
## .. customer_type = col_character(),
## .. adr = col_double(),
## .. required_car_parking_spaces = col_double(),
## .. total_of_special_requests = col_double(),
## .. reservation_status = col_character(),
## .. reservation_status_date = col_date(format = "")
## .. )
## - attr(*, "problems")=<externalptr>
```

To find out what columns we have in our data frame:

```
colnames(bookings_df)
```

```
## [1] "hotel" "is_canceled"
## [3] "lead_time" "arrival_date_year"
## [5] "arrival_date_month" "arrival_date_week_number"
## [7] "arrival_date_day_of_month" "stays_in_weekend_nights"
## [9] "stays_in_week_nights" "adults"
## [11] "children" "babies"
## [13] "meal" "country"
## [15] "market_segment" "distribution_channel"
## [17] "is_repeated_guest" "previous_cancellations"
## [19] "previous_bookings_not_canceled" "reserved_room_type"
## [21] "assigned_room_type" "booking_changes"
## [23] "deposit_type" "agent"
## [25] "company" "days_in_waiting_list"
## [27] "customer_type" "adr"
## [29] "required_car_parking_spaces" "total_of_special_requests"
## [31] "reservation_status" "reservation_status_date"
```

We want to create another data frame using `bookings_df` that focuses on the average daily rate, which is referred to as `adr` in the data frame, and `adults` :

```
new_df <- select(bookings_df, `adr`, adults)
```

To create new variables in our data frame, we can use the `mutate()` function. This will make changes to the data frame, but not to the original data set you imported. That source data will remain unchanged.

```
mutate(new_df, total = `adr` / adults)
```

```
## # A tibble: 119,390 × 3
##       adr adults total
##   <dbl>   <dbl> <dbl>
## 1     0         2     0
## 2     0         2     0
## 3    75         1    75
## 4    75         1    75
## 5    98         2    49
## 6    98         2    49
## 7   107         2   53.5
## 8   103         2   51.5
## 9    82         2    41
## 10  106.         2   52.8
## # i 119,380 more rows
```

Activity Wrap Up

We imported data using the `read_csv()` function, and worked with that data.